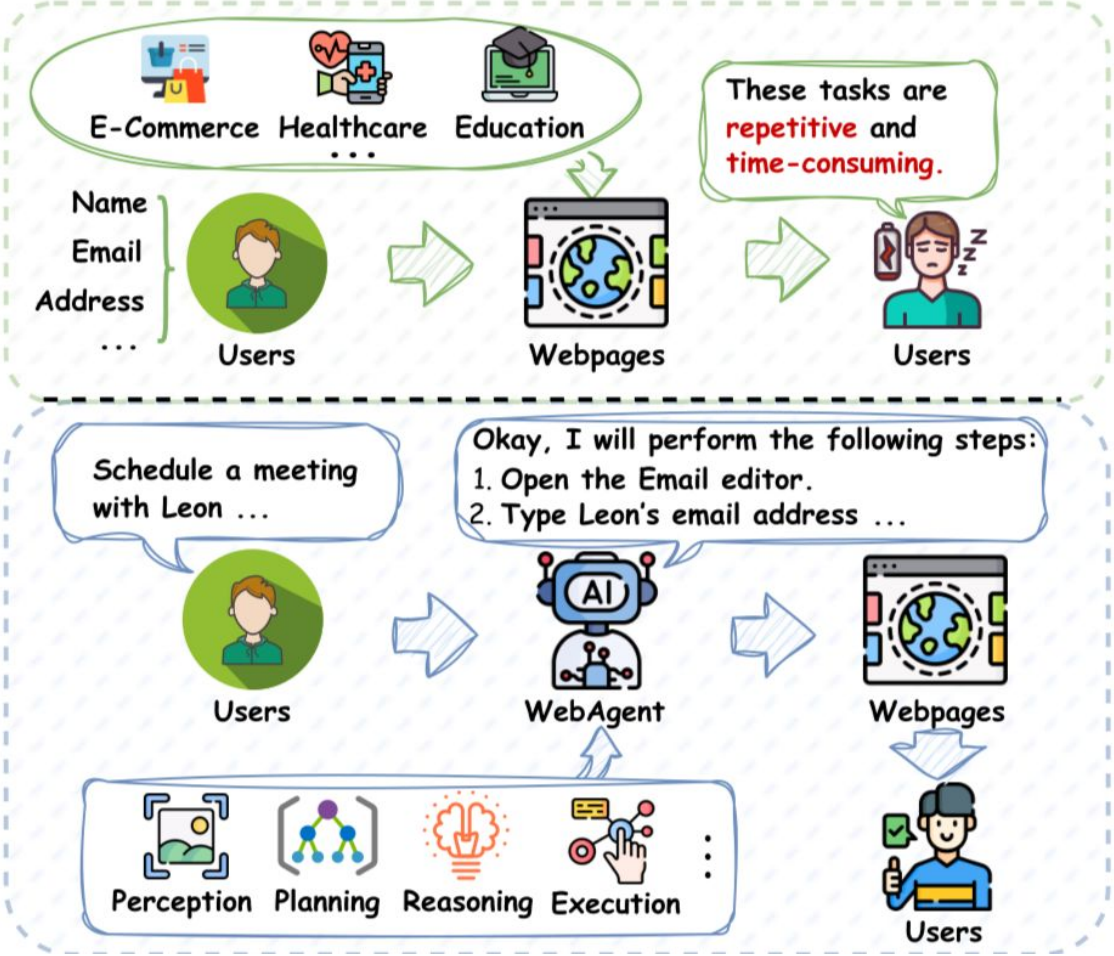
Web Agents

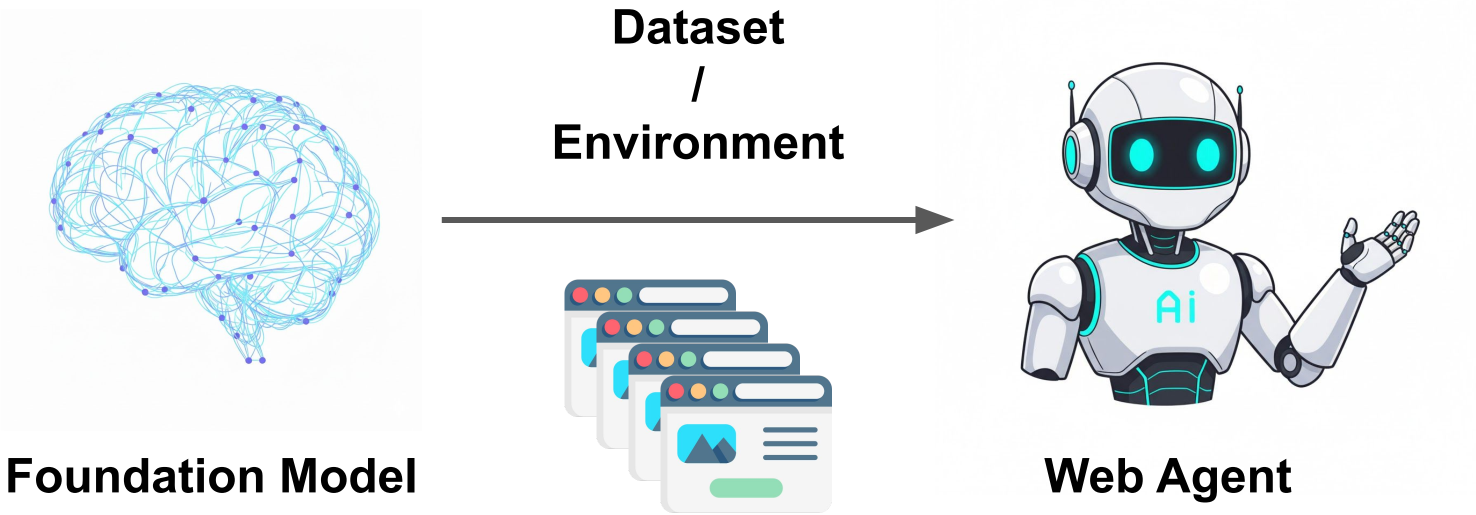
Yi-Chi Lee

10/21

Web Agents



https://arxiv.org/pdf/2503.23350



Dataset
/
Environment
Foundation Model
Ai
Web Agent

Timeline

- Word of Bits (2017)
- WebShop (2022)
- ChatGPT (2022)
- Mind2Web (2023)
- WebArena (2024)

World of Bits: An Open-Domain Platform for Web-Based Agents

Tianlin (Tim) Shi [1 2] Andrej Karpathy [2] Linxi (Jim) Fan [1] Jonathan Hernandez [2] Percy Liang [1]

https://dl.acm.org/doi/10.5555/3305890.3306005 (ICML '17)

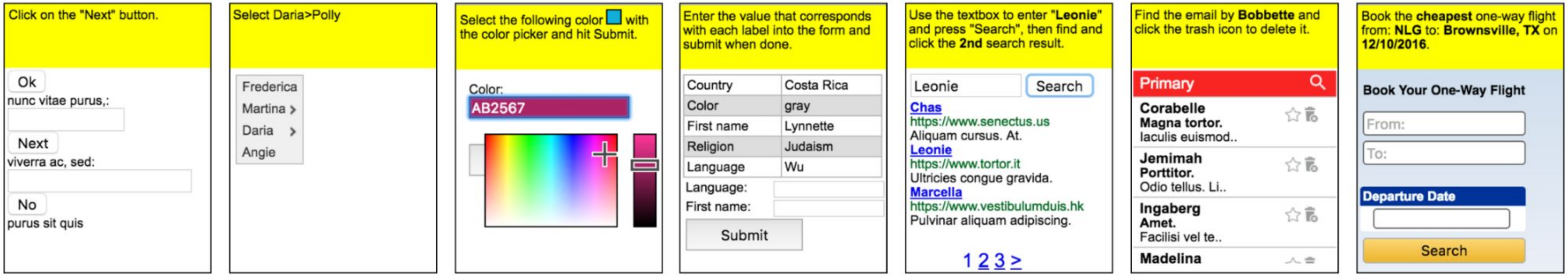
Word of Bits (WoB)

Interactive environment for reinforcement learning (OpenAI Gym interface)

- Observation
  - Raw screen pixels
  - Text DOM
- Action (low-level)
  - KeyEvent
  - PointerEvent
- Reward

MiniWoB

- Simple & straightforward web tasks
- Interactive Component: buttons, text fields, sliders, date pickers, etc.
- Reward: -1.0 (failure) ~ 1.0 (success)
- Success Rate (SR): ∑ 1[R > 0] / ∑ 1[R != 0]

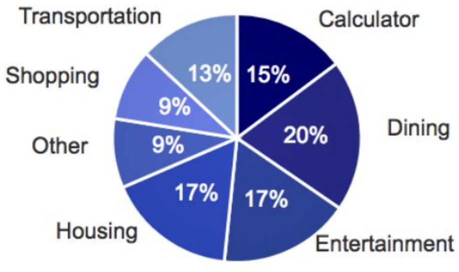


FormWoB

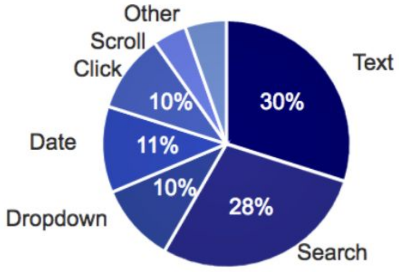
- **Offline Approximation**
  - Collect human demonstrations
  - Use a **proxy** for interaction between the agent and websites.
- Task: Flight booking websites
  - {from: DEN, to: JFK}
  - Reward: the fraction of key-value pairs that match those in human demonstrations

QAWoB

- More complex tasks



(a) Website categories.

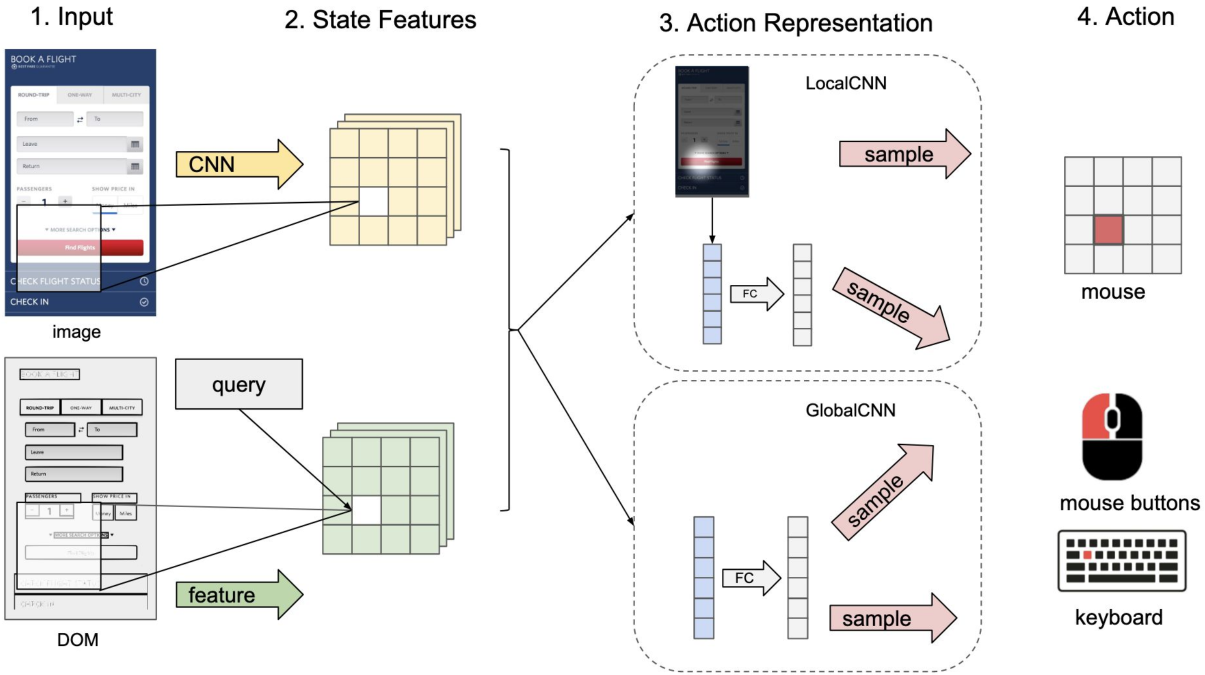


(b) GUI operations.

- Query Templates for scalability and diversity
  e.g., *"What is the cheapest restaurant that serves {{type of food}} near {{geographic location}}?"*
- Discourage queries that require too much reading comprehension

Agent Models

- State space
  - **Color Image**, DOM, and the query
- Action space
  - Cursor position
  - Mouse actions
  - Key actions
- Model Architecture
  - CNN
- Methods
  - Behavior cloning
  - RL (A3C)



Results on MiniWoB

	Random	SL	SL+RL	#envs
Success Rate (SR)	20.8	24.8	34.8	100
% Solved	12	17	26	100
SR: Click	28.1	35.2	48.5	57
SR: Drag	21.5	24.7	34.6	15
SR: Mouse+Keyboard	6.4	3.5	7.8	21
SR: Compound	3.3	3.8	4.3	7

Results on FormWoB

- treating the keyboard as categorical distribution was challenging
- achieves 20%–30% of human level performance on training queries, and 16% on test queries

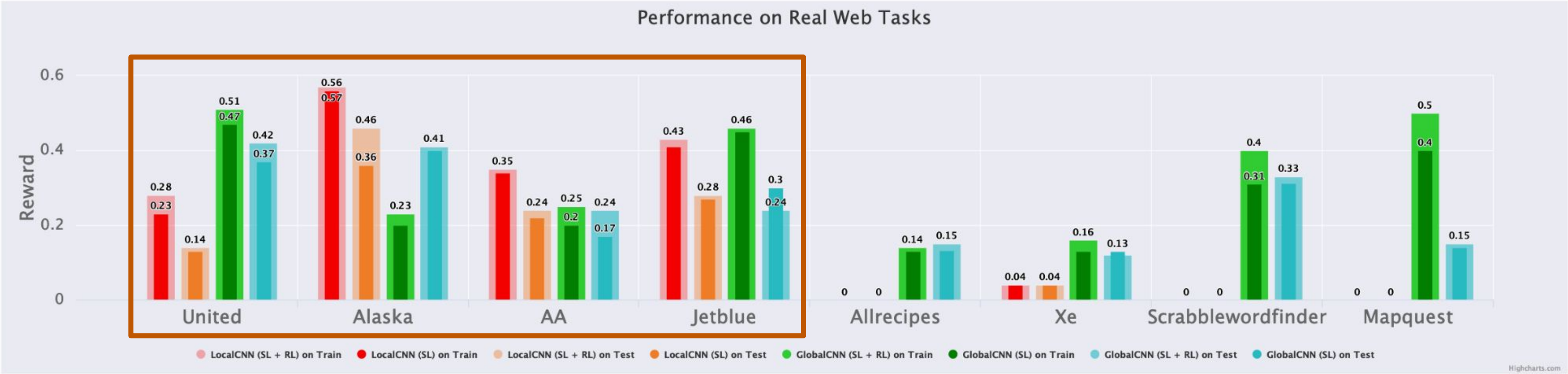


Figure 9. Rewards for the 4 FormWoB tasks and 4 QAWoB tasks. Random achieves 0, human is 1.

Results on QAWoB

- LocalCNN is incapable of fitting noisy human demonstrations

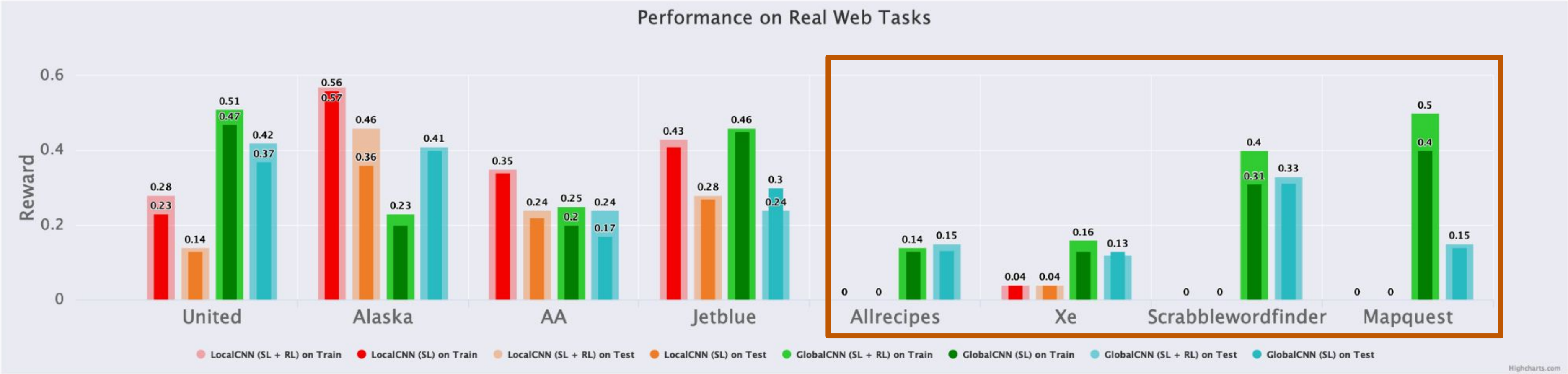


Figure 9. Rewards for the 4 FormWoB tasks and 4 QAWoB tasks. Random achieves 0, human is 1.

Can agents complete tasks across multiple webpages?

WebShop: Towards Scalable Real-World Web Interaction with Grounded Language Agents

Shunyu Yao* **Howard Chen*** **John Yang** **Karthik Narasimhan**
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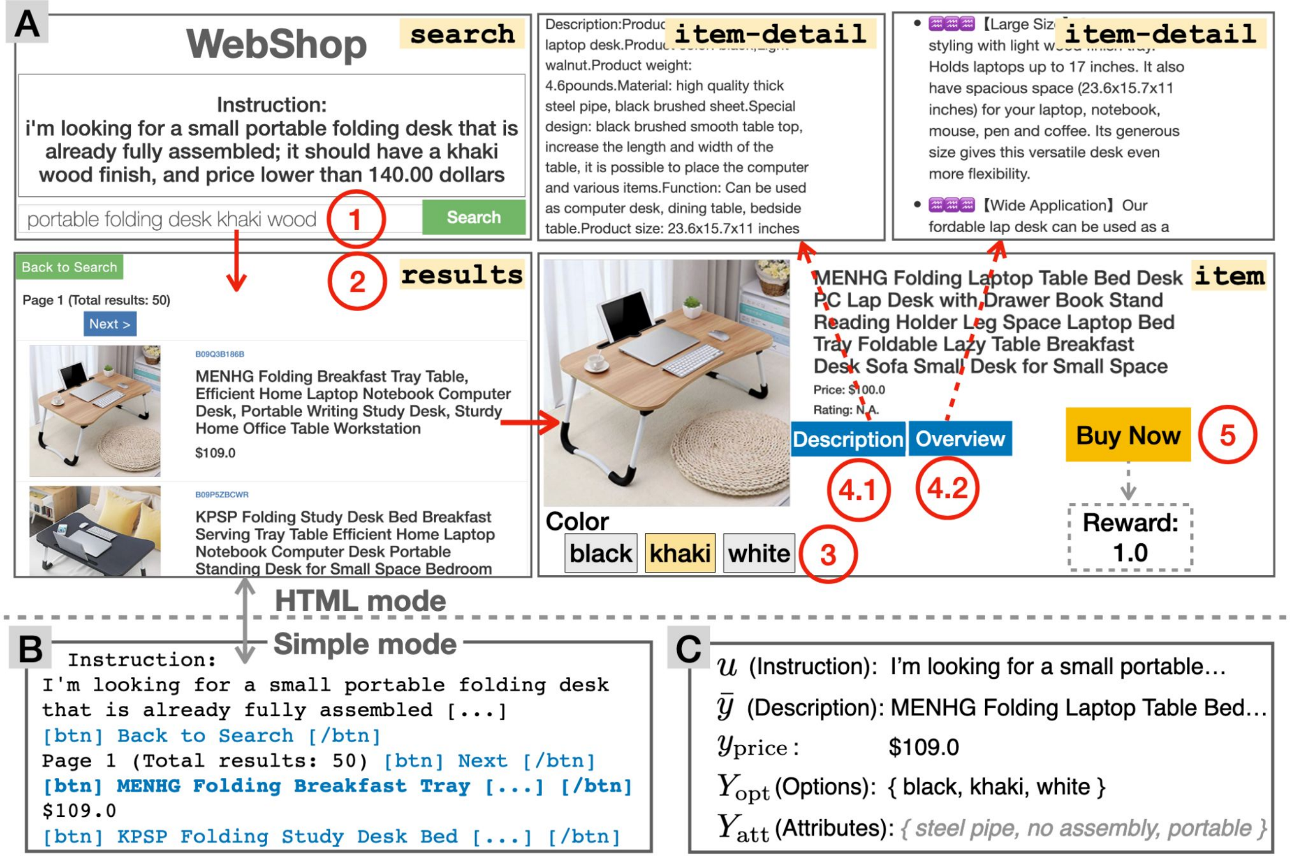
https://arxiv.org/pdf/2207.01206 (NeurIPS '22)

WebShop

A **simulated** e-commerce website environment



Task Formulation

- State: *search* page, *results* page, *item* page, *item-detail* page
- Action: search[Red shoes], choose[Size 9]
- Observation: HTML mode, simple mode
- Instruction: attributes + options + price
- Reward:

$$r = r_{\text{type}} \cdot \frac{|U_{\text{att}} \cap Y_{\text{att}}| + |U_{\text{opt}} \cap Y_{\text{opt}}| + \mathbf{1}[y_{\text{price}} \leq u_{\text{price}}]}{|U_{\text{att}}| + |U_{\text{opt}}| + 1}$$

Methods

- Rule Baseline

  Search with instruction, choose and buy the first item in the result page.
- Imitation Learning (IL)

  Search (BART): sequence-to-sequence text-generation: instruction → action

  Choose (BERT): from all the available click actions A(o) in observation o

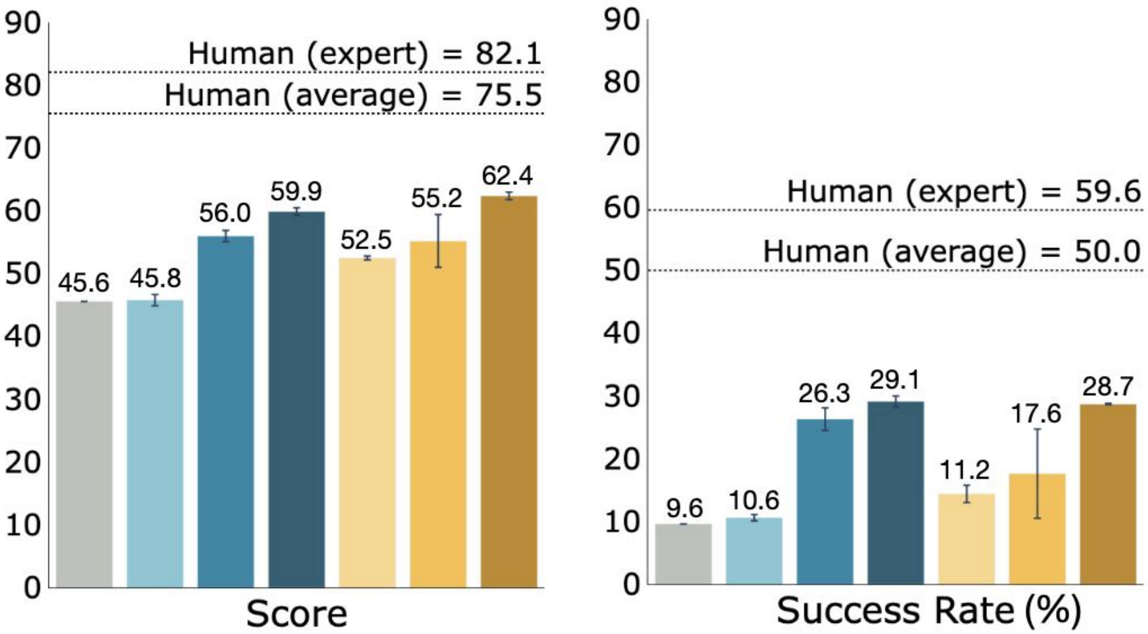
  Images are preprocessed by ResNet-50 and concatenated as observation o
- Reinforcement Learning (RL)

  Fine-tune the choice-based IL model

Results

- Task Score: 100 × avg. reward
- Success Rate (SR): the portion of instructions where r = 1

	LP Search	LP Choice	Human Demo	Use Reward
Rule				
IL w/o LP Choice	✓		✓	
IL w/o LP Search		✓	✓	
IL	✓	✓	✓	
RL	✓			✓
RL (RNN)				✓
IL+RL	✓	✓	✓	✓



Challenges from this benchmark

Instruction 1	Instruction 2
I want to find white blackout shades that are 66 inches in width and 66 inches in height. They need to be easy to install [...]	I need a gingko light and 20"x20" pillow cover that is hand painted [...]
Human Actions ($r = 1.0$, length $= 8$) `search` [66 inches in width and 66 inches in height white shades] `choose` [item : CALYX...] `choose` [Back to Search] `search` [66 x 66 blackout shades] `choose` [item : Milin...] `choose` [opt : 66"w x 66"h] `choose` [opt : cordless bottom up-blackout-white] `choose` [Buy]	**Human Actions** ($r = 1.0$, length $= 17$) `search` [gingko light 20"x20" pillow cover hand painted] `choose` [item : Maison...] [...] `choose` [Description] `choose` [< Previous] `choose` [Overview] `choose` [< Previous] [...] `choose` [item : Maison...] `choose` [opt : 20"x20"] `choose` [opt : nudes (gingko light)] `choose` [Buy]
IL+RL Actions ($r = 0.2$, length $= 3$) `search` [white blackout shades 65 inches in width and 66 inches in height] `choose` [item : Window...] `choose` [Buy]	**IL+RL Actions** ($r = 0.25$, length $= 3$) `search` [gingko light and 20x20 pillow cover hand painted] `choose` [item : UPOOS...] `choose` [Buy]

Challenges from this benchmark

- generation of good search queries and reformulation
- strategic exploration for navigating through the website
- robust language understanding for textual state and action spaces
- long-term memory for comparing items or backtracking

	Instr. text	IL BART	Human expert (first)	Human expert (last)
Score	94.9	94.5	94.5	**95.5**
Success Rate	85.4%	84.2%	85.6%	**87.8%**

Future Improvement

- robust search generation
- explicit memory modules
- better handling of noisy web text

MIND2WEB: Towards a Generalist Agent for the Web

Xiang Deng* **Yu Gu** **Boyuan Zheng** **Shijie Chen**
Samuel Stevens **Boshi Wang** **Huan Sun*** **Yu Su***
The Ohio State University
`https://osu-nlp-group.github.io/Mind2Web`

https://osu-nlp-group.github.io/Mind2Web/ (NeurIPS '23)

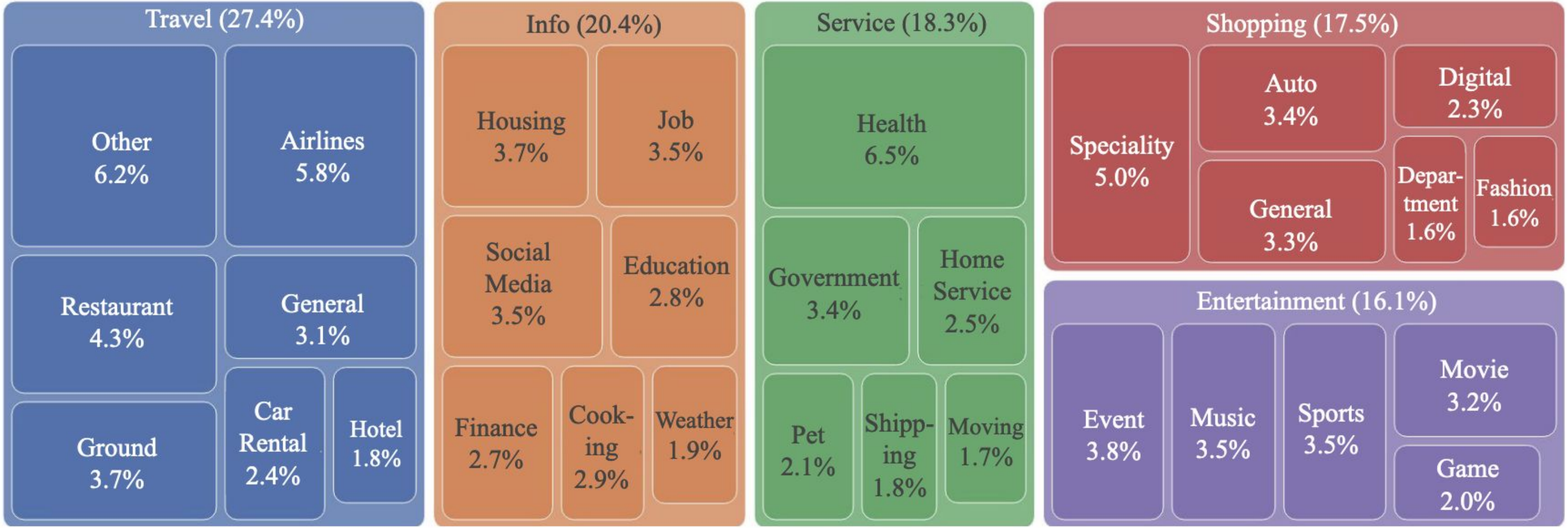
Mind2Web

- Real-world websites

  full traces of user interactions, webpage snapshots, and network traffic
- Diverse domain (31), websites (137), tasks (2000) and user interaction
- Enables researchers to conduct an initial exploration in using LLMs for building generalist web agents

Mind2Web

- Diverse domain (31), websites (137), tasks (2000) and user interaction



Instance in Mind2Web

1. High-level task description
2. Action Sequence (formed by multiple (Target Element, Operation) pairs)
    - Target Element: interactable element on current webpage
    - Click (including Hover and Press Enter), Type, and Select Option
3. Webpage Snapshot
    - Self-contained MHTML file
    - DOM snapshot
    - HAR file
    - Trace file

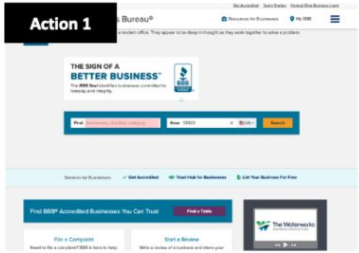
A Sample in Mind2Web

Task Description:
Show me the reviews for the auto repair business closest to 10002.

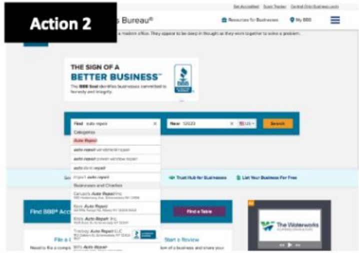
Action Sequence:

	Target Element	Operation
1.	[searchbox] *Find*	TYPE: ***auto repair***
2.	[button] *Auto Repair*	CLICK
3.	[textbox] *Near*	TYPE: ***10002***
4.	[button] *10002*	CLICK
5.	**[button] *Search***	**CLICK**
6.	[switch] *Show BBB Accredited only*	CLICK
7.	[svg]	CLICK
8.	[button] *Sort By*	CLICK
9.	**[link] *Fast Lane 24 Hour Auto Repair***	**CLICK**
10.	[link] *Read Reviews*	CLICK

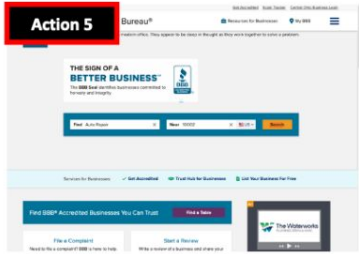
Webpage Snapshots:



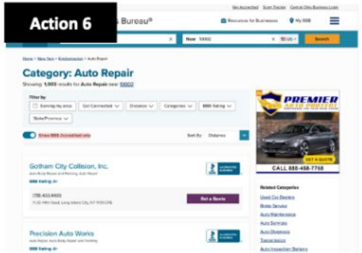
<input name="find_text" type="search">



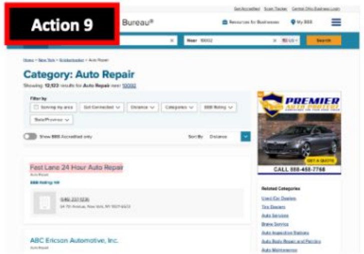
Auto Repair



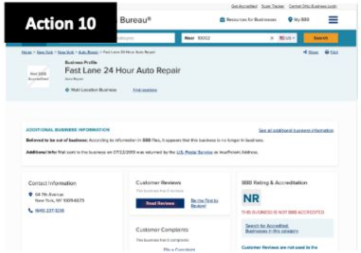
<button>Search</button>



<button>Show BBB Accredited only</button>



Fast Lane 24 Hour Auto Repair



Read Reviews

Data Collection

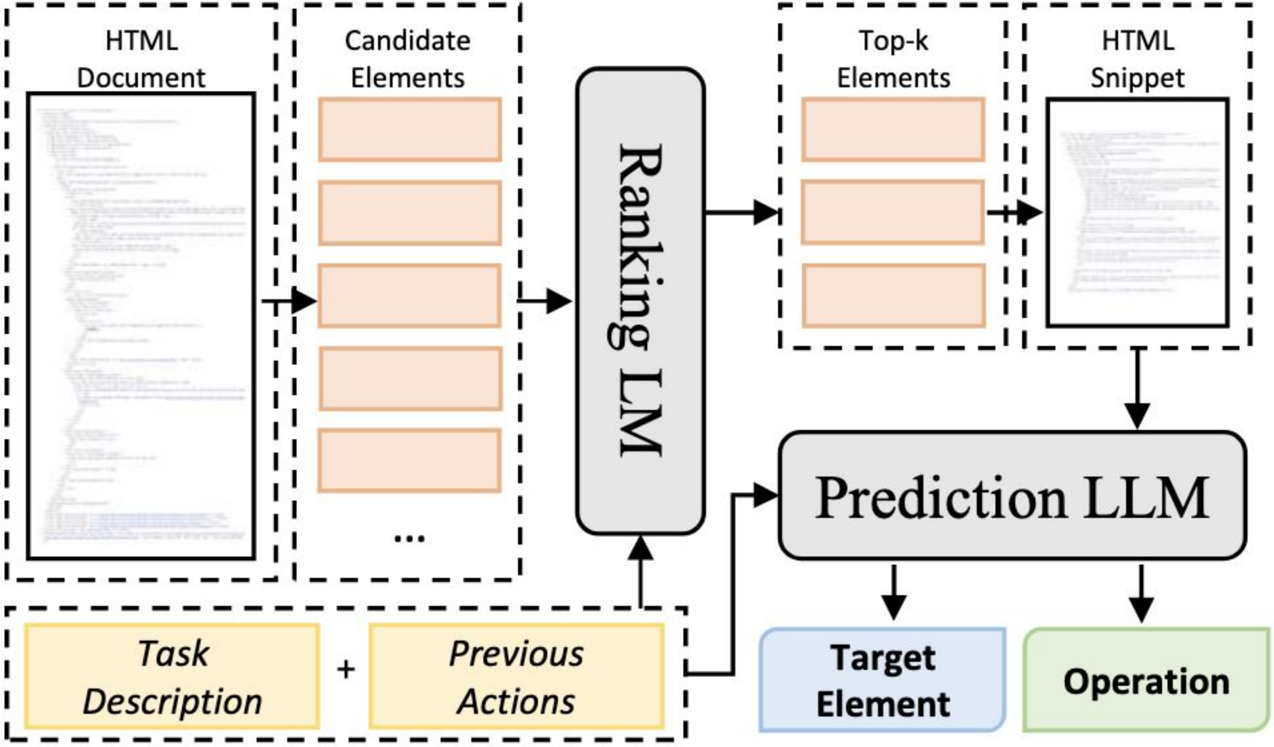
1. Select websites within each domain based on their popularity in the US
2. Task Criteria: diverse types, multiple rounds of interaction, and high-level goal
3. Task Demonstration
   - Element Selection
   - Operation Selection

MindAct

1. **Candidate Generation by Small LMs**
   - Input: task description, the snapshot of the webpage (at step t), and the actions (at step t -1)
   - Select top-k candidate DOM elements
2. **Action Prediction by LLMs**
   - Input: Snippets constructed by top-k candidates DOM elements and their neighbors
   - Multi-choice QA problem

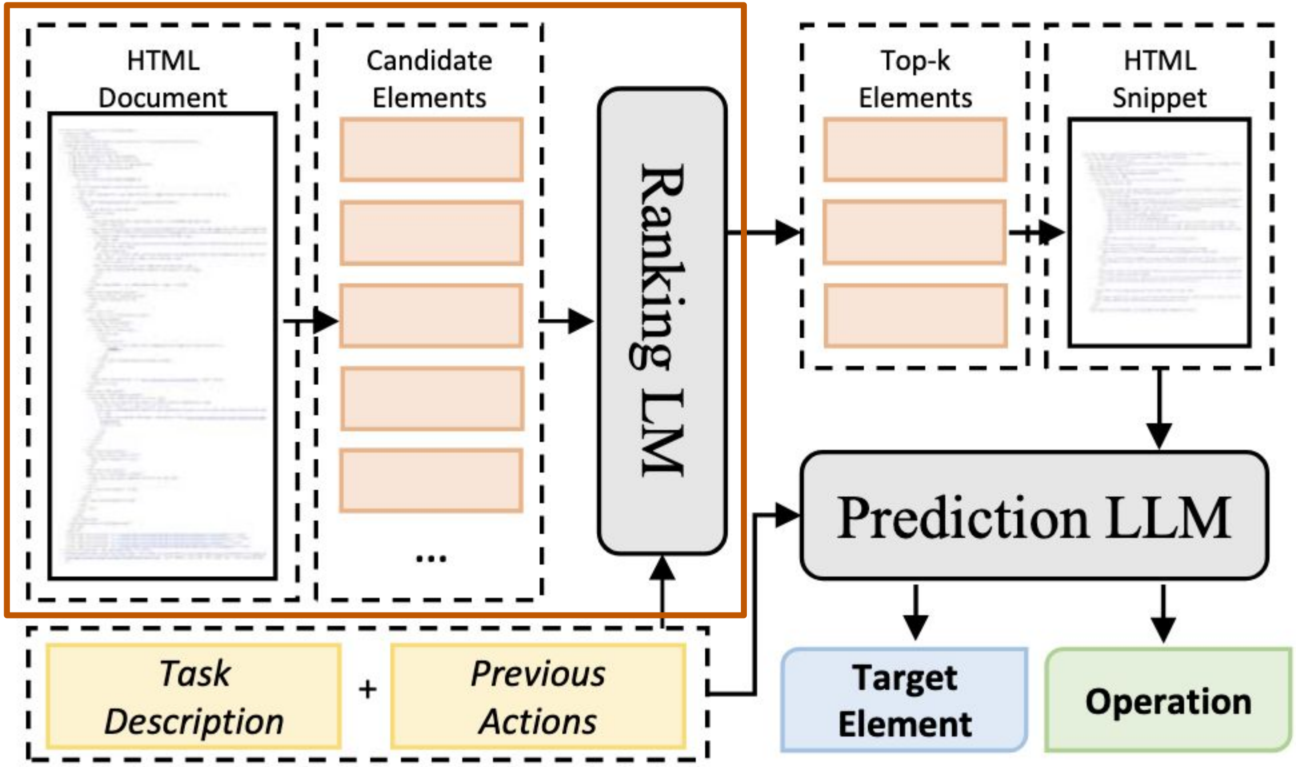
MindAct

Model Architecture



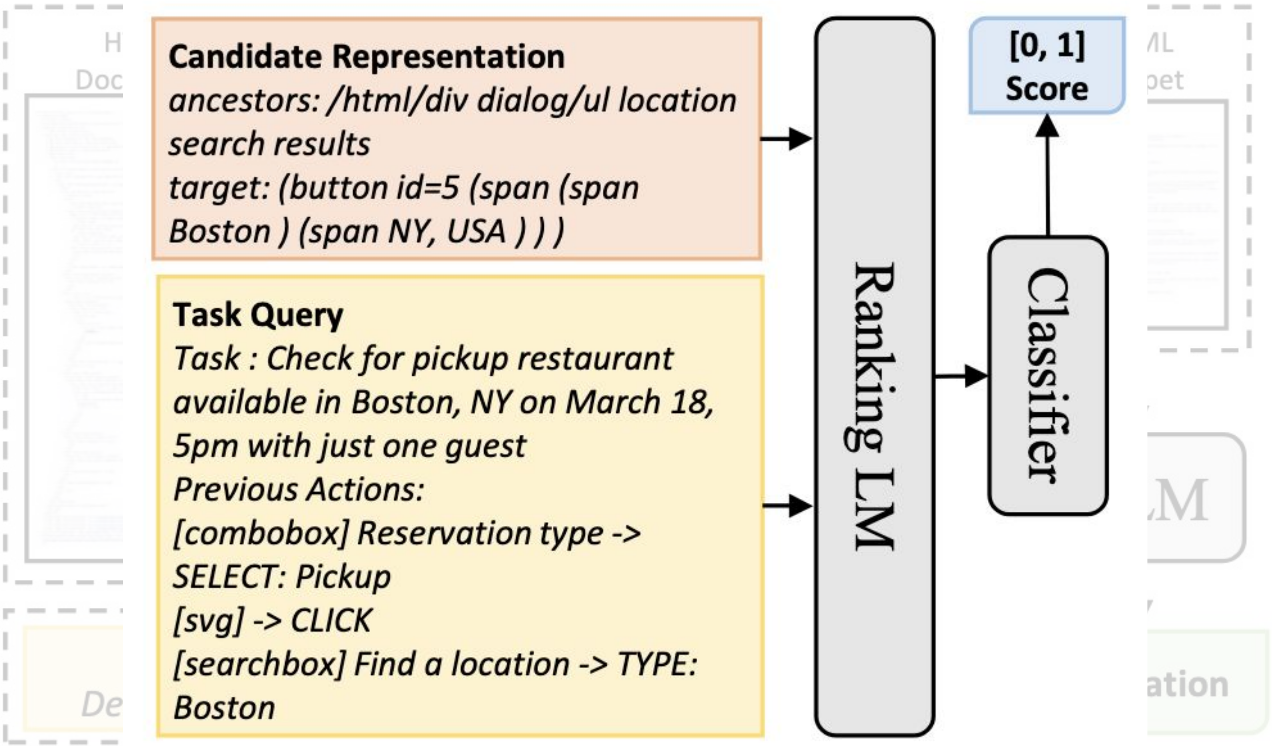
MindAct

Candidate Generation by Small LMs



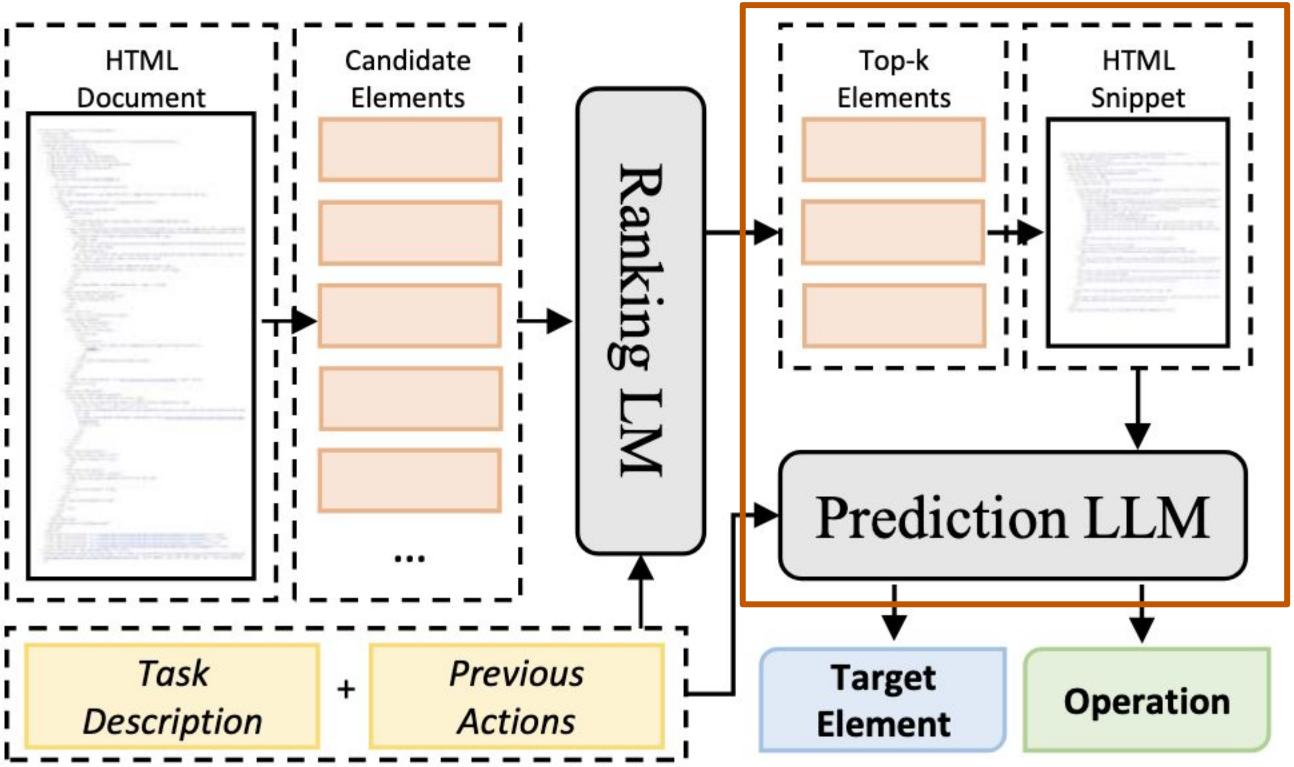
MindAct

Candidate Generation by Small LMs



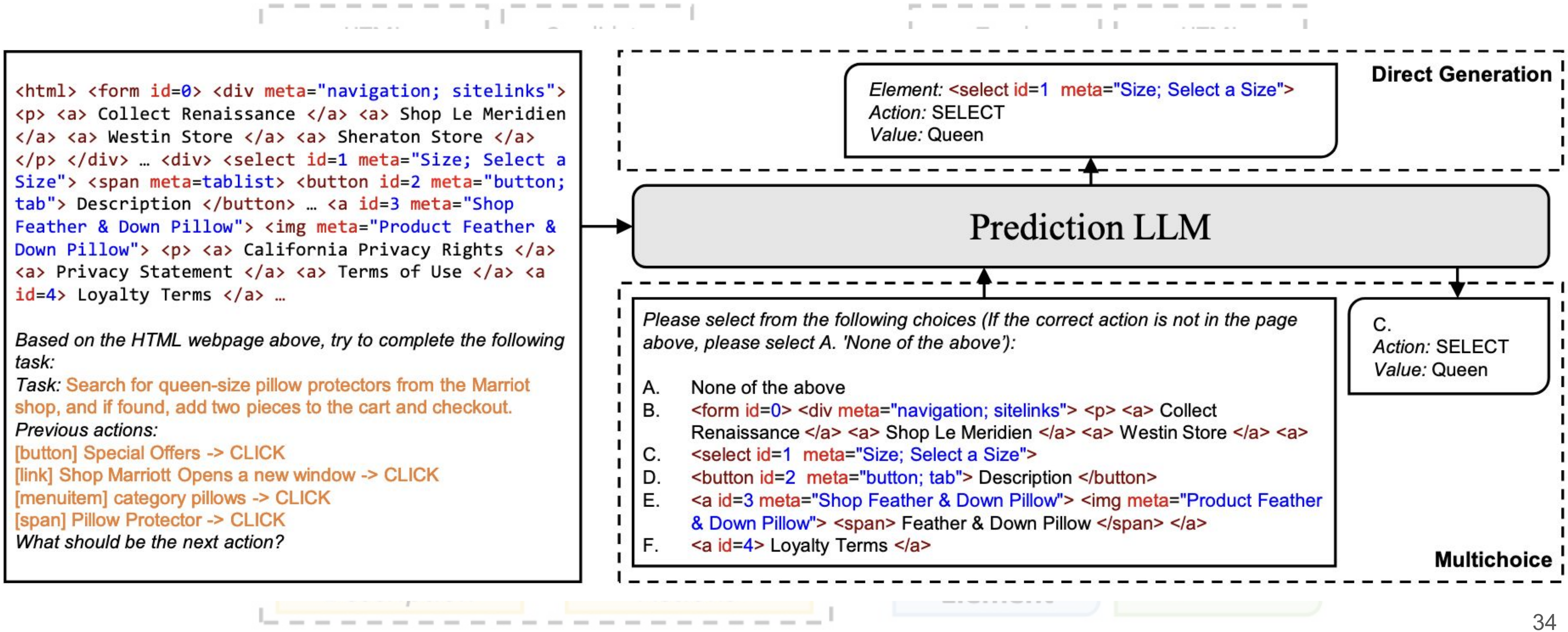
MindAct

Action Prediction by LLMs



MindAct

Candidate Generation by Small LMs



Results

$\text{Test}_{\text{Cross-Domain}}$: two top-level domains, Information and Service

$\text{Test}_{\text{Cross-Website}}$: 10 websites from each remaining top-level domain

$\text{Test}_{\text{Cross-Task}}$: randomly split 20% of the remaining data

Element Accuracy: compares the selected element with all acceptable elements

Operation F1: calculates token-level F1 score for the predicted operation

Step Success: if both the selected element and the predicted operation are correct

(Task) Success: if all steps have succeeded

Results

	Cross-Task				Cross-Website				Cross-Domain			
	Ele. Acc	**Op. F1**	**Step SR**	**SR**	**Ele. Acc**	**Op. F1**	**Step SR**	**SR**	**Ele. Acc**	**Op. F1**	**Step SR**	**SR**
Classification	26.8	–	–	–	21.6	–	–	–	24.5	–	–	–
Generation	20.2	52.0	17.5	0.0	13.9	44.7	11.0	0.0	14.2	44.7	11.9	0.4
MINDACT												
w/ Flan-$T5_B$	43.6	76.8	41.0	4.0	32.1	**67.6**	29.5	1.7	33.9	**67.3**	31.6	1.6
w/ Flan-$T5_L$	53.4	**75.7**	50.3	**7.1**	39.2	67.1	35.3	1.1	39.7	67.2	37.3	2.7
w/ Flan-$T5_{XL}$	**55.1**	**75.7**	**52.0**	5.2	**42.0**	65.2	**38.9**	**5.1**	**42.1**	66.5	**39.6**	**2.9**
w/ GPT-3.5	20.3	56.6	17.4	0.8	19.3	48.8	16.2	0.6	21.6	52.8	18.6	1.0
w/ GPT-4*	41.6	60.6	36.2	2.0	35.8	51.1	30.1	2.0	37.1	46.5	26.4	2.0

WEBARENA: A REALISTIC WEB ENVIRONMENT FOR BUILDING AUTONOMOUS AGENTS

Shuyan Zhou* **Frank F. Xu***
Hao Zhu † **Xuhui Zhou**† **Robert Lo**† **Abishek Sridhar**† **Xianyi Cheng**
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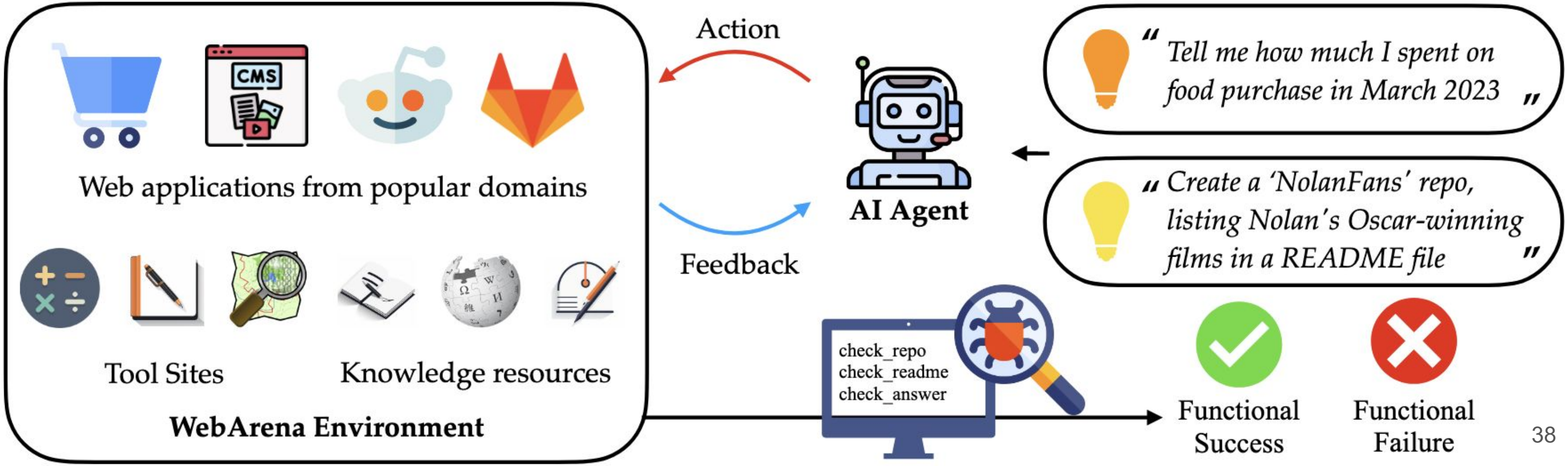
https://webarena.dev/ (ICLR '24)

WebArena

- **Fully operational**, **self-hosted** web applications
- High-level natural language task description
- Evaluate the **functional correctness**



Task Formulation $\mathcal{E} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T} \rangle$

- Input: natural language intent $\mathbf{i}$, the current observation $o_t \in \mathcal{O}$, the action history, and the observation history
- Output: action $a_t \in \mathcal{A}$
- Reward function: $r(\mathbf{a}_1^T, \mathbf{s}_1^T)$

- Observation: URL, opened tabs, and the page content of the focused tab
  - Page content: raw HTML, screenshot, accessibility tree
- Action: element operations, tab-related actions, and URL navigation actions
  - refer element by coordinates or unique ID

Intent Collection — 214 templates & 812 instances

Criteria:

1. abstract & high-level

   "post a greeting message on {{science}} subreddit"

2. creative

   *"create a Reddit account **identical to my GitLab one**"*

3. formulated as a template

   "create a {{site1}} account identical to my {{site2}} one"

Categories:

Information-seeking, Site navigation, and Content and configuration operation

Evaluation

Function	ID	Intent	Eval Implementation
$r_{\text{info}}(a^*, \hat{a})$	1	Tell me the name of the customer who has the most cancellations in the history	`exact_match`($\hat{a}$, "Samantha Jones")
	2	Find the customer name and email with phone number 8015551212	`must_include`($\hat{a}$, "Sean Miller") `must_include`($\hat{a}$, "sean@gmail.com")
	3	Compare walking and driving time from AMC Waterfront to Randyland	`fuzzy_match`($\hat{a}$, "walking: 2h58min") `fuzzy_match`($\hat{a}$, "driving: 21min")
$r_{\text{prog}}(\mathbf{s})$	4	Checkout merge requests assigned to me	`url=locate_current_url`($\mathbf{s}$) `exact_match`(URL, "gitlab.com/merge_requests?assignee_username=byteblaze")
	5	Post to ask "whether I need a car in NYC"	`url=locate_latest_post_url`($\mathbf{s}$) `body=locate_latest_post_body`($\mathbf{s}$) `must_include`(URL, "/f/nyc") `must_include`(body, "a car in NYC")

Experiment & Result

Context:

1. two examples
2. allowed action
3. overview of browser env

CoT	UA Hint	Model	SR	SR_{AC}	SR_{UA}
✓	✓	TEXT-BISON-001	5.05	4.00	27.78
✗	✓	GPT-3.5	6.41	4.90	38.89
✓	✓	GPT-3.5	8.75	6.44	58.33
✓	✓	GPT-4	11.70	8.63	**77.78**
✗	✗	GPT-3.5	5.10	4.90	8.33
✓	✗	GPT-3.5	6.16	6.06	8.33
✓	✗	GPT-4	**14.41**	**13.02**	44.44
-	✓	Human	78.24	77.30	100.00

Results

- Even when tasks are derived from the same template, they can present distinct challenges

Benchmark		Dynamic Interaction?	Realistic Environment?	Diverse Human Tasks?	Functional Correctness?
Mind2Web	(Deng et al., 2023)	✗	✓	✓	✗
Form/QAWoB	(Shi et al., 2017)	✗	✓	✓	✗
MiniWoB++	(Liu et al., 2018)	✓	✗	✗	✓
Webshop	(Yao et al., 2022a)	✓	✗	✗	✓
ALFRED	(Shridhar et al., 2020)	✓	✗	✗	✓
VirtualHome	(Puig et al., 2018)	✗	✗	✓	✗
AndroidEnv	(Toyama et al., 2021)	✓	✓	✗	✗
WebArena		✓	✓	✓	✓

Question?